# torch.diag#

https://pytorch.org/docs/stable/generated/torch.diag.html

**Description:**

If input is a vector (1-D tensor), then returns a 2-D square tensor with the elements of input as the diagonal. If input is a matrix (2-D tensor), then returns a 1-D tensor with the diagonal elements of input. The argument diagonal controls which diagonal to consider: If diagonal = 0, it is the main diagonal. If diagonal > 0, it is above the main diagonal.

## Function Signature

```
torch.diag(input, diagonal=0, *, out=None) → Tensor#
Parameters
Keyword Arguments
```

## Parameters

### Keyword

out (Tensor, optional) – the output tensor.

## Code Examples

```
>>> a = torch.randn(3)
>>> a
tensor([ 0.5950,-0.0872, 2.3298])
>>> torch.diag(a)
tensor([[ 0.5950, 0.0000, 0.0000],
        [ 0.0000,-0.0872, 0.0000],
        [ 0.0000, 0.0000, 2.3298]])
>>> torch.diag(a, 1)
tensor([[ 0.0000, 0.5950, 0.0000, 0.0000],
        [ 0.0000, 0.0000,-0.0872, 0.0000],
        [ 0.0000, 0.0000, 0.0000, 2.3298],
        [ 0.0000, 0.0000, 0.0000, 0.0000]])
```

```
>>> a = torch.randn(3, 3)
>>> a
tensor([[-0.4264, 0.0255,-0.1064],
        [ 0.8795,-0.2429, 0.1374],
        [ 0.1029,-0.6482,-1.6300]])
>>> torch.diag(a, 0)
tensor([-0.4264,-0.2429,-1.6300])
>>> torch.diag(a, 1)
tensor([ 0.0255, 0.1374])
```

## Notes & Warnings

**NOTE**

See also torch.diagonal() always returns the diagonal of its input. torch.diagflat() always constructs a tensor with diagonal elements specified by the input.

# Detailed Documentation

## Documentation

If input is a vector (1-D tensor), then returns a 2-D square tensor with the elements of input as the diagonal.

If input is a matrix (2-D tensor), then returns a 1-D tensor with the diagonal elements of input.

The argument diagonal controls which diagonal to consider:

If diagonal = 0, it is the main diagonal.

If diagonal > 0, it is above the main diagonal.

If diagonal < 0, it is below the main diagonal.

input (Tensor) – the input tensor.

diagonal (int, optional) – the diagonal to consider

out (Tensor, optional) – the output tensor.

torch.diagonal() always returns the diagonal of its input.

torch.diagflat() always constructs a tensor with diagonal elements specified by the input.

Get the square matrix where the input vector is the diagonal:

Get the k-th diagonal of a given matrix: